Worksheet 4.9

Chemical formulae

Answers

Covalent compounds

1. Complete the table by writing the chemical formula of the covalent compound formed from the listed elements.

Element 1	Element 2	Formula	Name of compound
Nitrogen	Chlorine	NCl_3	Nitrogen trichloride
Sulfur	Oxygen	SO_3	Sulfur trioxide
Carbon	Sulfur	CS_2	Carbon disulfide
Carbon	Oxygen	CO	Carbon monoxide
Phosphorus	Oxygen	P_2O_5	Diphosphorus pentoxide
Carbon	Fluorine	CF_4	Carbon tetrafluoride
Oxygen	Bromine	OBr_2	Oxygen dibromide
Nitrogen	Oxygen	N_2O_4	Dinitrogen tetroxide

Ionic compounds

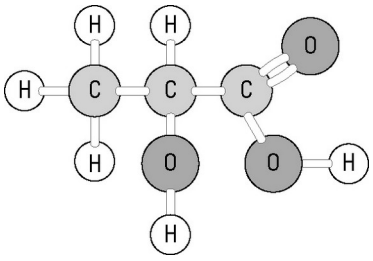
2. Use the table of valencies in your textbook to complete the table by writing the chemical formula of the ionic compound formed from the listed elements. Name each compound.

Positive ion	Negative ion	Formula	Name of compound
Silver	Iodide	AgI	Silver iodide
Lead	Sulfide	PbS	Lead sulfide
Calcium	Nitride	Ca_3N_2	Calcium nitride
Aluminium	Sulfide	Al_2S_3	Aluminium sulfide
Barium	Nitrate	$Ba(NO_3)_2$	Barium nitrate
Copper (II)	Sulfate	$CuSO_4$	Copper (II) sulfate
Sodium	Phosphate	Na_3PO_4	Sodium phosphate
Zinc	Hydrogen carbonate	$Zn(HCO_3)_2$	Zinc hydrogen carbonate

Structural formulae

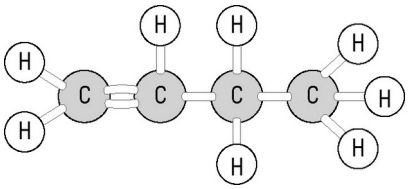
3. The structural formulae for various common compounds are shown. Write the molecular formula for each compound.

(a) Lactic acid



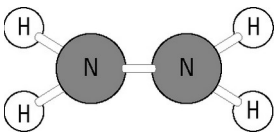
$C_3H_6O_3$

(b) Butene



C_4H_8

(c) Hydrazine



N_2H_4

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